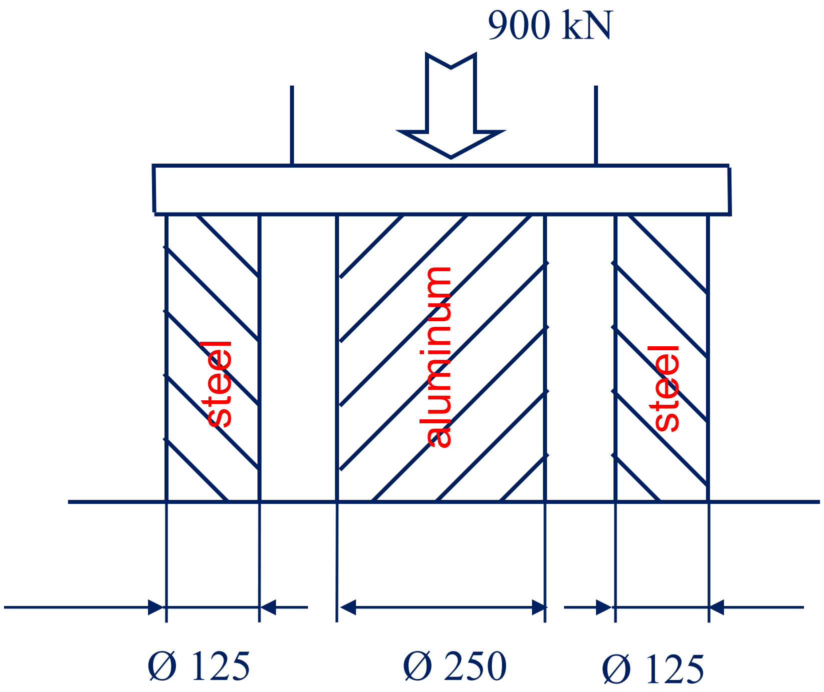
Fundamentals of Metal Forming Numerical Examples

Example 1

An aluminum cylinder ($D_{aluminum}$ = 250 mm) is placed on a non-deformable planar surface with two steel cylinders on the sides (D_{steel} = 125 mm).

Determine the compression stresses generated in the three parts subject to a load of 900 kN (non-deformable press). Suppose the materials remain in their elastic regions and there is no friction.



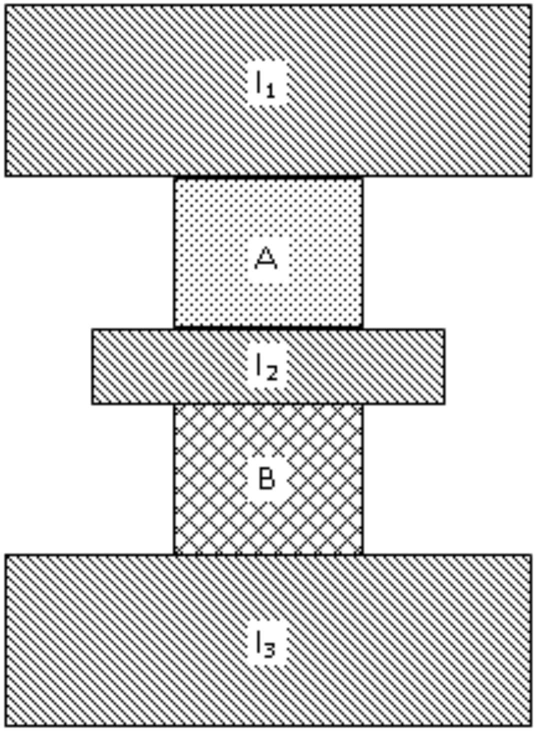
$E_{aluminum}$ = 70 000 N/mm^2

E_{steel} = 200 000 N/mm^2

Example 2

Consider a system consisting of three non-deformable bodies I1, I2, and I3, and two identical cylindrical bodies A and B (height 50 mm and diameter 40 mm), but A is made of steel (Y = 600 MPa) and B is made of bronze (Y = 320 MPa).

The body I2 is free to move in the vertical direction, the body I3 is fixed, and the body I1 is forced to lower by 10 mm.



Considering the behaviour of A and B as perfectly plastic and neglecting friction:

- Determine the final height of the two cylindrical bodies A and B;
- Determine the force necessary to obtain the 10 mm lowering of the body I1.

Example 3

A cylindrical sample in 1100-O aluminum alloy (k = 180 MPa, n = 0,20) with an initial height h_0 = 100 mm and an initial diameter d_0 = 80 mm, is compressed to a final height equal to h_2 = 40 mm in two steps between two flat dies with zero friction.

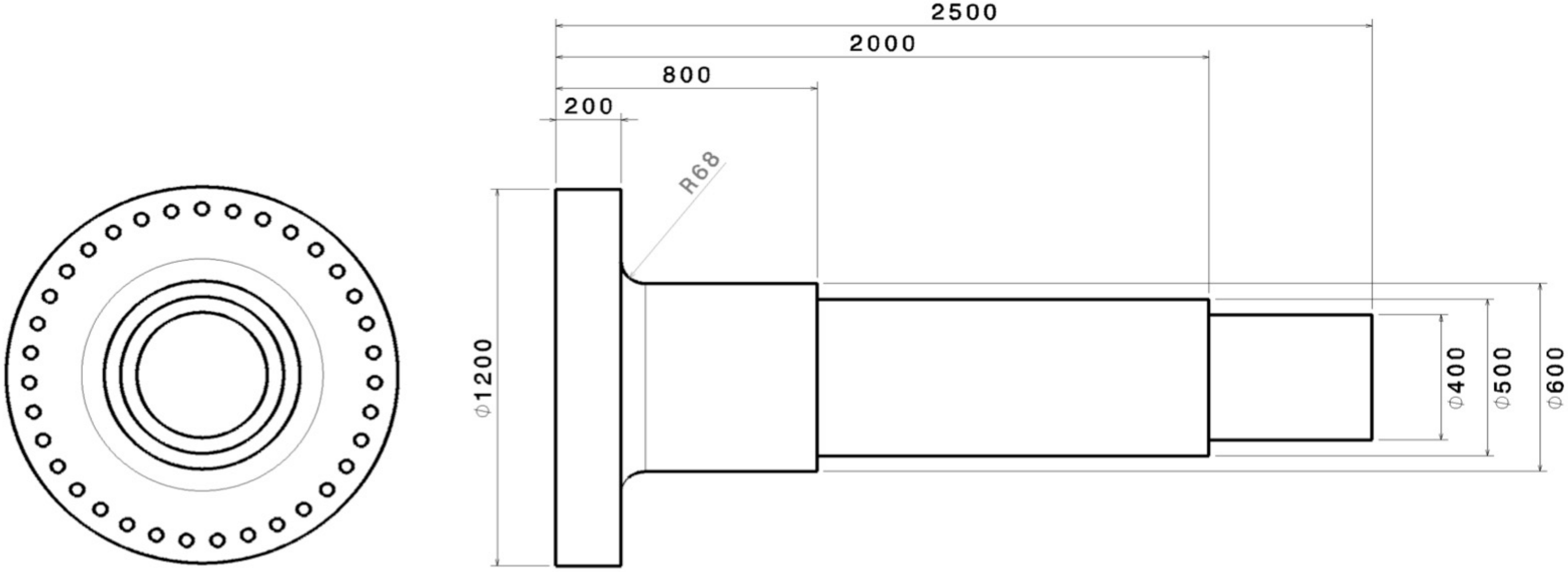
Determine:

- the engineering and true strains at the end of each step (h_1 = 70 mm, h_2 = 40 mm);
- the force required for each step;
- the strain energy density and the ideal work spent on the overall deformation.

Example 4

The production process of the shaft in the figure involves a set of forging operations and a subsequent upsetting of the flange. The process takes place at a temperature such as to guarantee a perfectly plastic behaviour with Y = 180 MPa.

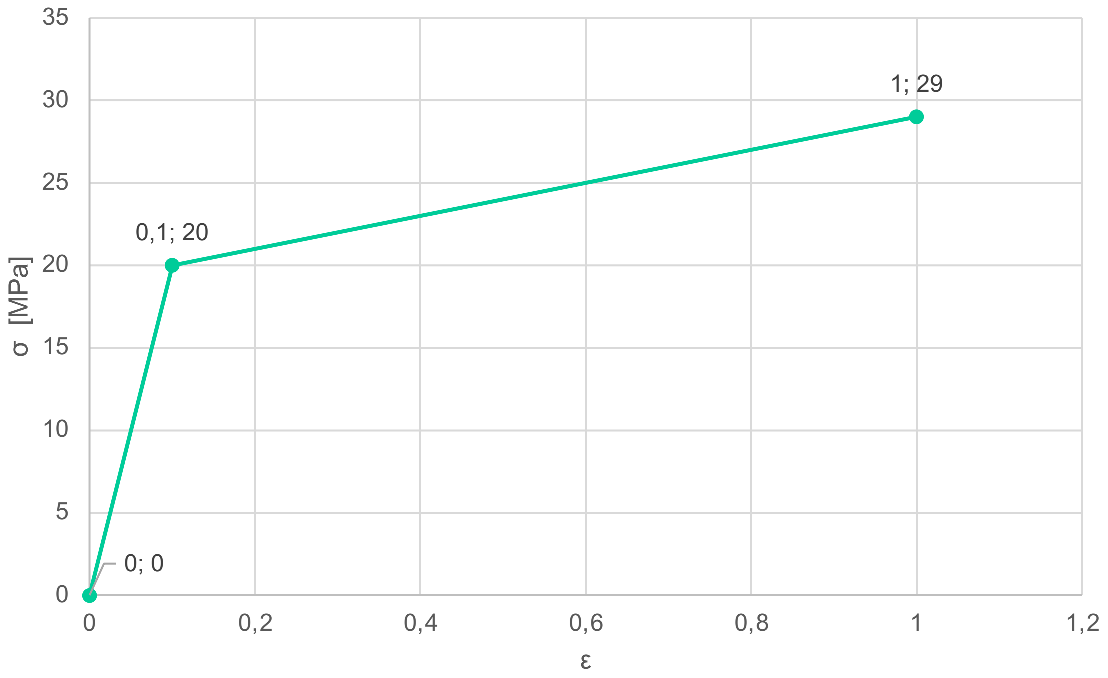
Assuming starting from a cylinder with a diameter of d_0 = 800 mm, determine the ideal work required to obtain each element of the shaft.



Example 5

A cylindrical specimen, with an initial height of 40 mm, is compressed with the aim of reaching a final height of 20 mm.

Considering the elastic recovery and the behaviour of the material as shown in the figure, what should be the actual height at which the specimen must be compressed?



Example 6

The height of an annealed copper cylinder with a diameter d_0 = 25 mm is reduced by 70% starting from a value h_0 = 50 mm by means of a metal forming process.

The material has a flow curve with parameters k = 315 MPa and n = 0,54.

Neglecting the barrelling phenomenon, determine / estimate:

- The maximum force in the absence of friction;
- The maximum force with friction equal to μ = 0,25;
- The maximum force with friction equal to μ=0,5;
- The strain energy density and the ideal work required to complete the forging.

Example 7

A cylindrical rivet in bronze C74500, with an initial diameter d_0 = 3 mm, is compressed to a final diameter equal to d_1 = 6 mm and a final height equal to h_1 = 3 mm.

The process takes place at a temperature such as to guarantee a perfectly plastic behaviour with Y = 170 MPa.

Assuming a friction coefficient $\mu = 0{,}2$, determine the required force.

Example 8

The height of an annealed carbon steel cylinder with a diameter d_0 = 50 mm is reduced by 20% starting from a value h_0 = 100 mm by open-die forging.

The material has a flow curve with parameters k = 530 MPa and n = 0,26, and UTS equal to 373 MPa.

Neglecting the barrelling phenomenon, determine / estimate:

- The maximum force with either no friction or friction equal to μ = 0,2;
- The strain energy density and the ideal work required to complete the forging;
- The minimum possible final hight.